

Moh Sabbir Saadat

Ph.D. - Computer Engineering
University of South Carolina

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HIGHLIGHTS

I am a researcher specializing in ubiquitous wireless sensing (mmWave radio) and digital health perception using multi-modal data. My expertise spans digital signal processing, deep learning, and machine learning. My work focuses on building robust, data-driven systems that bridge advances in AI/ML with intelligent perception applications. I also have experience in collaborative and inter-disciplinary research. **I am at the "All But Defense" stage in my Ph.D., and is looking for a suitable opportunity.**

EXPERIENCE

Graduate Research Assistant

SyReX Lab - University of South Carolina

January, 2019 — Present

Columbia, SC, USA

- Pursue novel ideas in ubiquitous wireless sensing and digital multi-modal health perception. (9 publications, 1 patent).
- Build synchronized multimodal systems (radio, camera, audio, smartwatch), execute data preprocessing, and develop evaluation benchmarks.
- Build complex ML models, and execute large-scale distributed training on GPU backend to benchmark them for various tasks.
- Spearhead inter-disciplinary research collaboration through knowledge-sharing and visualization of outcomes.
- Assisted a 400+ level class on computer networks with 100+ students for 6 semesters (Socket programming with Java, Python, and C).
- Assisted incoming graduate students through knowledge-sharing on research ideas, research culture, and work logistics.

Executive Engineer

Siemens Healthcare Limited

October, 2016 — November, 2018

Dhaka, Bangladesh

- Oversaw the technical requirements of potential clients in the medical imaging domain.
- Built liaison between the engineering department and the existing clientele.

EDUCATION

Ph.D. in Computer Engineering, University of South Carolina

May, 2026

M.Sc. in Computer Engineering, University of South Carolina

August, 2024

B.Sc. in Electrical & Electronics Engineering, Bangladesh University of Engineering & Technology

March, 2016

SKILLS

Software Engineering: Python, MATLAB, C++, Java, HTML, Git version control, Shell scripting, Linux.

ML libraries: PyTorch, TensorFlow, Keras, TensorBoard, Ray Tune, Scikit-learn, MATLAB Deep Learning.

ML models: Graph Neural Network, Vision Transformers, Generative Adversarial Network (GAN), Autoencoder, Few-shot Learning, LSTM.

Foundation AI models: MobileNet, ResNet, Mediapipe, HRNet, Whisper, WhisperX, BERT, Sentence-BERT.

Computational skills: Range-angle, Micro-doppler, Antenna array (FMCW radar), PCD processing, Clustering, STFT, Wavelet, DTW.

Tools: Latex, Gnuplot, Inkscape, Onshape CAD, TI mmWave Studio, ROS, Codex, Github Copilot.

PROJECTS

- **Vision and Audio-based digital health perception for post-stroke recovery assessment** (Multi-modal sensors, pose, ASR, time-series)
 - Built multi-sensors prototype with fine hardware-software synchronization. (4k camera, microphone array, smart-watch, wireless)

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- Facilitating the collection of clinical trials from stroke survivors and healthy subjects for post-stroke recovery assessment.
- Translating conventional clinical practices to a vision- and audio-based AI-enabled automated system.
 - * Assessing motor functionality from full-body images and facial abnormality from facial images. (*HRNet, Mediapipe*)
 - * Image processing pipeline to mask identity and lighting conditions in facial images.
 - * Audio noise cancellation and Audio Speech Recognition (ASR). (*Spectral noise cancellation, Whisper, WhisperX*)
 - * Cognitive assessments through Large Language Models (LLM). (*BERT, Sentence-BERT, BLIP*)
 - * Exploring speech intelligibility through phonetic comparison.
- **Co-existence of human-activity sensing on indoor wireless system** (*3D pose, Range-angle, μ -doppler, Graph Neural Network, Xformer*)
 - Graph neural network pipeline to overcome low-rate sensing signal due to co-existing networking.
 - Graph and Recurrent neural network (LSTM) to estimate sequences of 3D keypoints coordinates.
 - Deployment of activity recognition model on a real-time system ($\sim 98\%$ prediction accuracy with $\sim 2s$ initial latency).
 - Explored the Transformer model to develop an end-to-end system leveraging range-angle-doppler heatmaps sequence.
- **Imaging hidden objects with handheld millimeter-wave devices** (*SAR, Compressed sensing, cGAN, Ray-tracing*)
 - Leverage Synthetic Aperture Radar (SAR) principle to enable imaging from a small antenna array.
 - Overcome sparse sampling and motion non-linearity with signal processing methods. (*compressed sensing, unsupervised clustering*)
 - Overcome information loss in reflected wireless signal through cGAN-based image super-resolution.
 - Overcome the lack of ground truth data through ray-tracing based simulation of wireless signal reflections.
- **Temperature characterization and mitigation through scheduling** (*Mathematical modeling, Algorithm design*)
 - We study the temperature behavior of a 60-GHz mmWave access points.
 - We propose a smart scheduling scheme to overcome the overheating problem in mmWave networking devices.

RECENT PUBLICATIONS

- *[Under review]* **Moh Sabbir Saadat**, Ryan Titus, Haley Verkuilen, Phil Fleming, Souvik Sen, Sanjib Sur. “NeuroMotion: A Contactless and Automated Approach to Assessing Post-Stroke Severity in Facial and Motor Functions.” *IEEE Transactions on Neural Systems and Rehabilitation Engineering* [TNSRE 2026]
- *[To be submitted]* **Moh Sabbir Saadat**, Ryan Titus, Haley Verkuilen, Phil Fleming, Souvik Sen, Sanjib Sur. “NeuroConv: An Automated Approach to Assessing Post-Stroke Speech Functions from Audio.” *IEEE Transactions on Neural Systems and Rehabilitation Engineering* [TNSRE 2026]
- **Moh Sabbir Saadat**, Ryan Titus, Haley Verkuilen, Phil Fleming, Sanjib Sur, Souvik Sen. “A Contactless and Automated Approach to the Acute Stroke Assessment.” *AHA International Stroke Conference (February 2025)* [AHA ISC 2025]
- **Moh Sabbir Saadat**, Sanjib Sur. “Enabling Coexistence of Indoor Millimeter-Wave Networking and Human Activity Sensing.” *2024 IEEE/ACM International Conference on Connected Health: Applications, Systems, and Technologies (June 2024)* [CHASE 2024]

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- **Moh Sabbir Saadat**, Sanjib Sur. "Human Activity Sensing from Low-rate Samples under Integrated Networking." 2024 *IEEE/ACM International Conference on Connected Health: Applications, Systems, and Technologies (June 2024)* [CHASE 2024]
- **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "Aquila: Temperature-Aware Scheduler for Millimeter-Wave Devices and Networks." *The Elsevier High Confidence Computing journal* [Elsevier HCC 2024]
- Edward Sitar, **Moh Sabbir Saadat**, Sanjib Sur. "A Millimeter-Wave Wireless Sensing Approach for At-Home Exercise Recognition." *Proceedings of the ACM International Conference on Mobile Systems, Applications, and Services (June 2022)* [MobiSys 2022]
- Hem Regmi, **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "SquiggleMilli: Approximating SAR Imaging on Mobile Millimeter-Wave Devices." *Proceedings of the ACM on Interactive, Mobile, Wearable, and Ubiquitous Technologies (September 2021)* [IMWUT 2021]
- Hem Regmi, **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "ZigZagCam: Pushing the Limits of Hand-held Millimeter-Wave Imaging." *ACM International Workshop on Mobile Computing Systems and Applications (February 2021)* [HotMobile 2021] (Best Poster Runner-up Award)
- **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "A Case for Temperature-Aware Scheduler for Millimeter-Wave Devices and Networks." *The 28th IEEE International Conference on Network Protocols (October 2020)* [ICNP 2020]
- **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi. "Bringing Temperature-Awareness to Millimeter-Wave Networks." *ACM International Conference Mobile Computing and Networking (September 2020)* [MobiCom 2020]
- **Moh Sabbir Saadat**, Sanjib Sur, Srihari Nelakuditi, Parmesh Ramanathan. "MilliCam: Hand-held Millimeter-Wave Imaging." *The 29th IEEE Conference on Computer Communications and Networks (August 2020)* [ICCCN 2020]

PATENT

Sanjib Sur, Moh Sabbir Saadat, Srihari Nelakuditi, "Heat Dissipation for Millimeter-wave Devices with Antenna Switching" (Granted: February 2023)

AWARDS

- **Student travel grant, CHASE'24:** Presenting work on enabling sensing on integrated networking system
- **Best poster runner-up at ACM HotMobile'21:** For early work on hand-held millimeter-wave imaging
- **Saluting the Nation Builders of Tomorrow 2008/2010, The Daily Star :** For outstanding achievement in GCE O/A level

PROFESSIONAL SERVICES

- **External Reviewer**
 - IEEE Transactions on Mobile Computing 2022
 - IEEE/ACM Transactions on Networking 2023
 - IEEE International Conference on Mobility, Sensing, and Networking (MSN'23)
 - IEEE/ACM International Conference on Internet of Things Design and Implementation (IoTDI'23)
 - EAI Ubiquitous 2022
 - Pervasive and Mobile Computing 2025
- **Other Services**
 - Served as the Vice President of Bangladesh Student Association at the University of South Carolina 2021-2022
 - Served as the General Secretary of Bangladesh Student Association at the University of South Carolina 2020-2021
 - Served as an adjudicator at the NALSAR University Debating Tournament, Hyderabad, India 2012